H2: Electrostatics

2.1 The electric field

2.1.1 introduction

principle of superposition	= interaction between any two charges is completely unaffected by the presence of others > calculate the forces on the charges separately, then the total force F = F_1+F_2+…
electrostatics	= case where the source charges are stationary > the test charge may be moving

2.1.2 Coulombs law

Coulombs law	$$\mathbf{F} = \frac{1}{4\pi\epsilon_0}\frac{qQ}{\imath^2}\hat{\boldsymbol{\imath}}.$$ with ε_0 the permittivity of free space and r: $\boldsymbol{\imath} = \mathbf{r} - \mathbf{r}'$; the relative position

2.1.3 the electric field

force on a charge	for a test charge Q we have a total force, due to sources q_i: $$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \ldots = \frac{1}{4\pi\epsilon_0}\left(\frac{q_1 Q}{\imath_1^2}\hat{\boldsymbol{\imath}}_1 + \frac{q_2 Q}{\imath_2^2}\hat{\boldsymbol{\imath}}_2 + \ldots\right)$$ $$= \frac{Q}{4\pi\epsilon_0}\left(\frac{q_1}{\imath_1^2}\hat{\boldsymbol{\imath}}_1 + \frac{q_2}{\imath_2^2}\hat{\boldsymbol{\imath}}_2 + \frac{q_3}{\imath_3^2}\hat{\boldsymbol{\imath}}_3 + \ldots\right),$$
electric field	= field only dependent on source charges > tells us how a test charge of Q=1 will move through space: $$\mathbf{E}(\mathbf{r}) \equiv \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{q_i}{\imath_i^2}\hat{\boldsymbol{\imath}}_i.$$ defined via: $\mathbf{F} = Q\mathbf{E}$,
properties of the electric field	1: E is a vector field dependent on **r** the position in space > however the formula is written in terms of relative position $\boldsymbol{\imath}_i$ 2: its independent of the test charge Q

2.1.4 continuous charge distributions

E for continuous charge	If our charge is continuously distributed, we can take the integral instead: $$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int\frac{1}{\imath^2}\hat{\boldsymbol{\imath}}\,dq.$$ now we make distinctions between the types of distributions - line with λ charge per unit length: dq = λdl' with dl' an element of length along the line - surface with σ charge per unit area: dq = σda' with da' an element of area on the surface - volume with ρ charge per unit volume: dq = ρdτ' with dτ' an element of volume >> we then have: - $\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int\frac{\lambda(\mathbf{r}')}{\imath^2}\hat{\boldsymbol{\imath}}\,dl';$ - $\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int\frac{\sigma(\mathbf{r}')}{\imath^2}\hat{\boldsymbol{\imath}}\,da';$ - $\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int\frac{\rho(\mathbf{r}')}{\imath^2}\hat{\boldsymbol{\imath}}\,d\tau'.$ >> Coulomb's law

2.2 divergence and curl of electrostatic fields

2.2.1 field lines, flux and Gauss's law

flux of E

the flux of E through a surface S is:

$$\Phi_E \equiv \int_S \mathbf{E} \cdot d\mathbf{a},$$

Gauss's law

for one charge at the origin we have, through a sphere of radius r:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \int \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r^2}\hat{\mathbf{r}}\right) \cdot (r^2 \sin\theta \, d\theta \, d\phi \, \hat{\mathbf{r}}) = \frac{1}{\epsilon_0} q.$$

now take a bunch of point charges, the flux through the sphere is:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \sum_{i=1}^{n} \left(\oint \mathbf{E}_i \cdot d\mathbf{a} \right) = \sum_{i=1}^{n} \left(\frac{1}{\epsilon_0} q_i \right)$$

or thus, Gauss's law for any closed surface S:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}},$$

differential Gauss's law

Apply the divergence theorem to Gauss's law:

$$\oint_{\mathcal{S}} \mathbf{E} \cdot d\mathbf{a} = \int_{\mathcal{V}} (\nabla \cdot \mathbf{E}) \, d\tau.$$

and Q_{enc} can be written in terms of charge density ρ:

$$Q_{\text{enc}} = \int_{\mathcal{V}} \rho \, d\tau.$$

we then have:

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho.$$

2.2.2 divergence of E

divergence of E

The differential form of Gauss's law is in agreement with the previous section, nl:

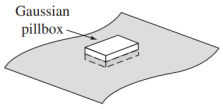
$$\nabla \cdot \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \nabla \cdot \left(\frac{\hat{\boldsymbol{\imath}}}{\imath^2} \right) \rho(\mathbf{r}') \, d\tau'.$$

with: $\nabla \cdot \left(\frac{\hat{\boldsymbol{\imath}}}{\imath^2} \right) = 4\pi\delta^3(\boldsymbol{\imath})$. => $\nabla \cdot \mathbf{E} = \frac{1}{4\pi\epsilon_0} \int 4\pi\delta^3(\mathbf{r} - \mathbf{r}')\rho(\mathbf{r}') \, d\tau' = \frac{1}{\epsilon_0}\rho(\mathbf{r}),$

2.2.3 application of Gauss's law

importance symmetries

Symmetries in the problem can lead you to choose a different approach:
1: Spherical symmetry: make your Gaussian surface a concentric sphere
2: Cylindrical symmetry: make your Gaussian surface a coaxial cylinder
3: Plane symmetry: make your Gaussian surface a pillbox



2.2.4 the curl of E

curl of E

take a point charge at the origin: $\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\mathbf{r}}.$

the line integral of the field from some points a to b is: $\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l}.$

or spherical $d\mathbf{l} = dr\,\hat{\mathbf{r}} + r\,d\theta\,\hat{\boldsymbol{\theta}} + r\sin\theta\,d\phi\,\hat{\boldsymbol{\phi}}$, :

$$\mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}dr.$$

thus we have:

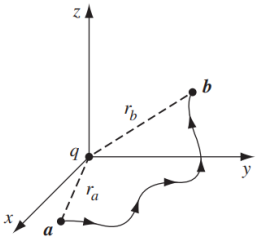
$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0}\int_{\mathbf{a}}^{\mathbf{b}} \frac{q}{r^2}dr = \frac{-1}{4\pi\epsilon_0}\frac{q}{r}\Big|_{r_a}^{r_b} = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r_a} - \frac{q}{r_b}\right)$$

therefore, on a closed path we have:

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0,$$

and via Stokes theorem:

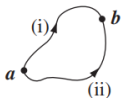
$$\nabla \times \mathbf{E} = \mathbf{0}.$$

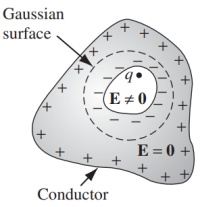


for multiple fields, we have superposition, thus:

$$\nabla \times \mathbf{E} = \nabla \times (\mathbf{E}_1 + \mathbf{E}_2 + \ldots) = (\nabla \times \mathbf{E}_1) + (\nabla \times \mathbf{E}_2) + \ldots = \mathbf{0}.$$

>> thus, this applies for any configuration of source charges

2.3 electric potential	
2.3.1 introduction to potential	
electric potential V	We know that ∇xE=0, thus any closed line integral of E will result in zero: $\oint \mathbf{E}\cdot d\mathbf{l}=0$ > the integral from point a to b is always the same, regardless the chosen path > we can define a scalar: $V(\mathbf{r}) \equiv -\int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E}\cdot d\mathbf{l}.$ 
potential difference	Now we have: $V(\mathbf{b})-V(\mathbf{a}) = -\int_{\mathcal{O}}^{\mathbf{b}} \mathbf{E}\cdot d\mathbf{l} + \int_{\mathcal{O}}^{\mathbf{a}} \mathbf{E}\cdot d\mathbf{l}$ $= -\int_{\mathcal{O}}^{\mathbf{b}} \mathbf{E}\cdot d\mathbf{l} - \int_{\mathbf{a}}^{\mathcal{O}} \mathbf{E}\cdot d\mathbf{l} = -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E}\cdot d\mathbf{l}.$
derivative of V	The fundamental theorem of gradients states: $V(\mathbf{b})-V(\mathbf{a}) = \int_{\mathbf{a}}^{\mathbf{b}} (\nabla V)\cdot d\mathbf{l},$ so we have: $\int_{\mathbf{a}}^{\mathbf{b}} (\nabla V)\cdot d\mathbf{l} = -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E}\cdot d\mathbf{l}.$ or: $\mathbf{E} = -\nabla V.$
2.3.2 comments on potential	
superposition of V	just like F and E, V also adhere's to the principle of superposition: $V = V_1 + V_2 + \ldots$
2.3.3 Poisson's equation and Laplace's equation	
Poisson's equation	We know ∇E = ρ/ε₀, thus we have: $\nabla^2 V = -\frac{\rho}{\epsilon_0}.$
Laplace's equation	in regions with no charge, ie ρ=0, we have: ∇²V=0
2.3.4 the potential of a localized charge distribution	
V in terms of ρ > inverted Poisson's eq.	For a point charge we have an electric field E: $\mathbf{E}\cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\,dr.$ for a reference point at infinity, the potential is: $V(r) = -\int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E}\cdot d\mathbf{l} = \frac{-1}{4\pi\epsilon_0}\int_{\infty}^{r}\frac{q}{r'^2}\,dr' = \frac{1}{4\pi\epsilon_0}\frac{q}{r'}\Big|_{\infty}^{r} = \frac{1}{4\pi\epsilon_0}\frac{q}{r}.$ due to superposition, we have for multiple charges: $V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\mathbf{r}')}{\mathscr{r}}\,d\tau'.$ > for line and surface charges we have: $V = \frac{1}{4\pi\epsilon_0}\int \frac{\lambda(\mathbf{r}')}{\mathscr{r}}\,dl' \quad \text{and} \quad V = \frac{1}{4\pi\epsilon_0}\int \frac{\sigma(\mathbf{r}')}{\mathscr{r}}\,da'.$
2.4 work and energy in electrostatics	
2.4.1 the work it takes to move a charge	
work to move a charge	The work required to move a point charge from a to b is: $W = \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F}\cdot d\mathbf{l} = -Q\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E}\cdot d\mathbf{l} = Q[V(\mathbf{b})-V(\mathbf{a})].$ now, if we take a point at infinity and move a charge from there to a position r, we have: $W = Q[V(\mathbf{r})-V(\infty)],$ but V(inf)=0 thus we have: $W = QV(\mathbf{r}).$

2.4.2 the energy of a point charge distribution	
work to create a charged system	The work to assemble a configuration of point charges is: $$W = \frac{1}{2}\sum_{i=1}^{n} q_i V(\mathbf{r}_i).$$
2.4.3 energy of a continuous charge distribution	
energy of charge distribution	for a volume charge density ρ we have: $$W = \frac{1}{2}\int \rho V \, d\tau.$$ we use Gauss's law to get: $$\rho = \epsilon_0 \nabla \cdot \mathbf{E}, \quad \text{so} \quad W = \frac{\epsilon_0}{2}\int (\nabla \cdot \mathbf{E}) V \, d\tau.$$ integrate by parts to find: $$W = \frac{\epsilon_0}{2}\left[-\int \mathbf{E} \cdot (\nabla V)\, d\tau + \oint V\mathbf{E} \cdot d\mathbf{a}\right].$$ and since E=-∇V: $$W = \frac{\epsilon_0}{2}\left(\int_{\mathcal{V}} E^2 \, d\tau + \oint_{\mathcal{S}} V\mathbf{E} \cdot d\mathbf{a}\right)$$ if we integrate over all space, the surface integral becomes 0 and we find: $$W = \frac{\epsilon_0}{2}\int E^2 \, d\tau$$ OVER ALL SPACE
2.4.4 comments on electrostatic energy	
superposition principle of W	W doesn't adhere to the superposition principle > the energy of the system != the sum of the energies of its parts ex: $$W_{\text{tot}} = \frac{\epsilon_0}{2}\int E^2 \, d\tau = \frac{\epsilon_0}{2}\int (\mathbf{E}_1 + \mathbf{E}_2)^2 \, d\tau$$ $$= \frac{\epsilon_0}{2}\int \left(E_1^2 + E_2^2 + 2\mathbf{E}_1 \cdot \mathbf{E}_2\right) d\tau$$ $$= W_1 + W_2 + \epsilon_0 \int \mathbf{E}_1 \cdot \mathbf{E}_2 \, d\tau.$$ >> there are cross-terms
2.5 conductors	
2.5.1 basic properties	
properties of conductors	1: E=0 in a conductor nl: if there were a field, there would be flowing charges, thus no electro*statics* 2: ρ=0 inside a conductor > via Gauss's law ∇.E=ρ/ϵ_0 3: any net charges resides on the surface 4: a conductor is an equipotential > for two points **a** and **b,** V(**a**) = V(**b**) since $V(\mathbf{b}) - V(\mathbf{a}) = -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} = 0,$ 5: E is perpendicular to the surfaces, just outside the conductor
2.5.2 induced charges	
induced charge	hold a charge +q to a uncharged spherical conductor > the two will attract eachother nl: q will attract negative charges in the conductor and repel the pos charges > there will be a net force of attraction
Faraday cage	place a charge in a cavity inside the conductor > in the cavity there will be an electric field > in the conductor and surroundings there will be no electric field nl: the other charges cancel out the electric field 

2.5.3 surface charge and the force on a conductor	
electric field of a conductor	The field outside a conductor is: $$\mathbf{E} = \frac{\sigma}{\epsilon_0}\hat{\mathbf{n}},$$ perpendicular to the surface > the potential is: $\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$
force on a conductor	In the presence of an electric field, any surface charge will experience a force > the force per unit area is then **f** = σ**E** > however, the electric field is then discontinuous at a surface charge > we don't know whether we have to use E_{above} or E_{under} > we need to average the two: $$\mathbf{f} = \sigma \mathbf{E}_{\text{average}} = \frac{1}{2}\sigma(\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}}).$$ proof: consider a small patch of surface surrounding the point in question > the total field consists of two parts: $$\mathbf{E} = \mathbf{E}_{\text{patch}} + \mathbf{E}_{\text{other}}$$ the patch however cannot exert a force on itself > the force is exclusively due to the E_{other} and this suffers no discontinuity > the discontinuity is due to the charge on the patch, putting a field $\sigma/2\varepsilon_0$ on each side > thus: $\mathbf{E}_{\text{above}} = \mathbf{E}_{\text{other}} + \frac{\sigma}{2\epsilon_0}\hat{\mathbf{n}}$, $\mathbf{E}_{\text{below}} = \mathbf{E}_{\text{other}} - \frac{\sigma}{2\epsilon_0}\hat{\mathbf{n}}$, hence: $\mathbf{E}_{\text{other}} = \frac{1}{2}(\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}}) = \mathbf{E}_{\text{average}}$.
electrostatic pressure	for a conductor the force per unit area is: $$\mathbf{f} = \frac{1}{2\epsilon_0}\sigma^2\hat{\mathbf{n}}.$$ this amounts to an outward electrostatic pressure on the surface > we can express this as: $$P = \frac{\epsilon_0}{2}E^2.$$
2.5.4 capacitors	
capacitor	suppose we have two conductors with charge +Q and -Q opposite to eachother > V is cte over a conductor, we have a potential difference: $$V = V_+ - V_- = -\int_{(-)}^{(+)} \mathbf{E}\cdot d\mathbf{l}.$$ E is proportional to Q, given by Coulomb's law: $$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\int \frac{\rho}{\mathcal{r}^2}\hat{\boldsymbol{\mathcal{r}}}\, d\tau,$$
capacitance	E is proportional to Q and V > their proportion is constant for a certain arrangement: $$C \equiv \frac{Q}{V}.$$ >> entirely determined by the sizes, chapes and separation of the two conductors